CS|W-23|CS3T002|AI3T002|02

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

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(An Autonomous Institute, with NAAC "A" Grade)

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Education to Eternity

॥ज्ञानं सर्वार्थ साधनम्॥

B. Tech ~~Summer~~ / Winter – 2024

(ESE / ~~Make-up~~ / ~~Supplementary~~) Examination

Branch: Computer Science & Engineering/Artificial Intelligence Sem:- III

Subject: Operating System · Subject Code: CS3T002/AI3T002

Marks:100 Date of Exam: 16/11/2024 Time: 3hrs

Instructions:

1. **Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.**
2. **All the questions carry marks as indicated.**
3. **Assume suitable data whenever necessary.**
4. **Illustrate your answer whenever necessary with the help of neat sketches.**
5. **Use of Programmable Calculator is prohibited.**
6. **Don't use red pen for writing the answers.**

COs: Students should be able to:

1. Understand the basics of how operating systems work.
2. Explain how processes and CPU scheduling function in an operating system.
3. Solve common process synchronization problems.
4. Describe memory management concepts, including virtual memory.
5. Comprehend disk management and the role of file systems in an operating system.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one/two word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define a process.	L1/CO2	2
Q.2	State the purpose of a bootstrap program.	L1/CO1	2
Q.3	Identify the main components of computer system organization.	L2/CO1	2
Q.4	List out the types of threads	L1/CO2	2
Q.5	Rewrite the critical section problem.	L2/CO3	2
Q.6	Define contiguous memory allocation.	L1/CO4	2
Q.7	Summarize virtual memory and its purpose in an operating system.	L1/CO5	2
Q.8	State demand paging and Page fault.	L1/CO4	2
Q.9	Give example internal fragmentation.	L2/CO4	2
Q.10	Define race condition.	L1/CO3	2
Q.11	Distinguish between logical and physical addresses.	L2/CO4	2
Q.12	State how the available resources are calculated in Banker's algorithm?	L1/CO3	2

SECTION B

1. **All Questions are compulsory.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.13	Consider the following processes with their burst times:	L5/CO2	4

Process	Burst Time
P1	6
P2	8
P3	7
P4	3

Assume all processes arrive at time 0. Calculate the waiting time and turnaround time using Shortest Job First (SJF) algorithm

Q.14	Illustrate the process state transitions in details.	L3/CO2	4
Q.15	Discuss the services provided by an operating system and their significance to users and applications.	L2/CO4	4
Q.16	Describe the C-SCAN disk scheduling algorithm and how it differs from the SCAN algorithm.	L2/CO5	4
Q.17	Compare and contrast internal and external fragmentation, including their causes and solutions.	L2/CO3	4
Q.18	Explain the reader-writer problem and its challenges	L2/CO3	4

SECTION C

1. **Solve any FOUR Questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.19	Discuss the methods for handling deadlocks.	L2/CO3	7
Q.20	Illustrate the concepts of fixed partitioning and variable partitioning in memory allocation	L4/CO4	7
Q.21	Analyse the advantages and disadvantages of contiguous memory allocation compared to paging	L4/CO4	7
Q.22	Explain the FIFO page replacement algorithm and its basic principle.	L5/CO5	7
Q.23	Describe the types of CPU schedulers.	L6/CO2	7
Q.24	Compare the First-Come, First-Served (FCFS) and Shortest Remaining Time First (SRTF) scheduling algorithms, highlighting their performance characteristics.	L4/CO2	7

SECTION D

1. **Solve any TWO Questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.25	Consider a system with 3 frames in memory. The following sequence of page references occurs: **Page Reference String**: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 7, 0, 1, 2. Determine the number of page faults that occur using the following page replacement algorithms: **a)** FIFO (First-In, First-Out) **b)** LRU (Least Recently Used) **c)** Optimal	L3/CO4	12
Q.26	Analyse the characteristics of various types of operating systems, including real-time, batch, and time-sharing systems	L4/CO5	12
Q.27	Consider a system that contains five processes P1, P2, P3, P4, P5 and the three resource types A, B and C. Following are the resources types: A has 10, B has 5 and the resource type C has 7 instances.	L5/CO3	12

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P1	0	1	0	7	5	3	3	3	2
P2	2	0	0	3	2	2			
P3	3	0	2	9	0	2			
P4	2	1	1	2	2	2			
P5	0	0	2	4	3	3			

Evaluate the following using the banker's algorithm:

a) Need matrix?
b) Determine if the system is safe or not.
c) Safe Sequence